Conditional machine-level distinct branch invariant

The completed UInt64 expression and weighted field-validity results compose into one theorem. For prime $5 \le p < 2^{31}$, two canonical degree-one inputs, four nonzero word weights, common normalized Mumford validity and distinct normalized x, the modeled source guard selects distinct. Its literal degree-two payload is canonical, has nonzero word weights and preserves normalized Mumford validity.

Composition. Canonical bounds convert nonzero input words into nonzero field weights. The six-coordinate decoding theorem identifies the literal payload with the field formula. The weighted field theorem supplies output validity and nonzero field weights; rewriting gives normalized word validity, and decoded nonzero implies word nonzero. No coefficient identities are assumed.

Control. Over F7 use nonunit-weight $a = (0, 0, 2, 3)$ and $b = (2, 3, 3, 2)$, normalized points $(0, 0), (1, 1)$ on $f = X^2$. Both input divisibilities are constructed and the full theorem is applied. This is polynomial algebra, not a smooth genus-two fixture. Equal normalized x and zero input z boundary checks are retained.

Checks. Main theorem and separate axiom readback pass; controls pass after explicit field-residue rewriting. Failed controls using decide on noncomputable division and insufficient norm_num reduction and an intermediate modulus-instance mismatch are preserved. Only propext, Classical.choice and Quot.sound occur.

Scope. Degree-one plus degree-one to degree-two support, arbitrary common polynomial f. Source conditions remain Field.n=1, tables=None and ordinary execution/counter headroom. No compiler/counter, other-branch, divisor-class or whole-walk claim. No scientific data or costs. Independent review pending; source hashes and receipts are retained.